

Midterm Project 1: Linear Classification of Heart Disease Patients

BME 395/495

Due: Sunday, April 18th at 12:00 am CT

1 Introduction to Jupyter Notebook

Your project template is given as a **Jupyter notebook**. Jupyter notebook is a Python environment that allows editing and running notebook documents via a web browser. Using Jupyter notebook, your assignment can be conveniently divided into parts and you can follow the instructions on the notebook. You are expected to submit your project as a full Jupyter notebook.

To start a Jupyter Notebook (.ipynb) file:

1. Open your Anaconda navigator. It should be installed as an app on your computer. (If not, go to <https://www.anaconda.com/products/individual> and install Anaconda navigator.)
2. Click **Launch** on Jupyter Notebook. This should open a new tab on your default browser with the name **Home Page** and the URL <http://localhost:8888>. On the page, you should see the list of the files on your default Python directory.
3. Click on **New** on upper right or left corner and on **Python 3** for a new notebook file.

You should have a blank notebook file on a new tab. The space where you can write your code is called a cell. You can run a cell after writing your code by clicking on the black arrow on the left. After that cell is run, you can insert a new cell from **Insert** menu above. You can rename your notebook file and save your work from **File**. (You can use the shortcuts as well.)

2 Project Template

After setting up your environment, you can download your `MidtermProject1_Template.ipynb` file on Canvas and move the file to your default Python directory. After that, the file should appear in the list on your **Home Page** tab. Click on it to open the file on a new tab.

3 Project

In this first midterm project, you need to implement a Logistic Regression model that linearly classifies the patients with and without a heart disease. The dataset is already introduced in the lecture.

As explained in the headers of the notebook file, the first thing you should do is to import the necessary libraries and functions. After that, you need to import and preprocess the dataset. For the Logistic Regression model, you need to write six functions in total: five helper functions and one main function for Logistic Regression.

Three helper functions, `initialize`, `sigmoid` and `predict`, need to be written from scratch. `Initialize` and `predict` were a part of the Linear Regression code we have written in class but you did not see them. `Sigmoid` is necessary to implement the classification algorithm, so we did not have it in the regression code. Please follow the instructions on the notebook file to write them and run the given test cases to check if the functions are running properly.

The other two helper functions, `calculate_cost_gradient` and `GradientDescent`, will be similar to the functions we have written for Linear Regression. There are a few differences:

- The cost function is not MSE anymore, it is crossentropy.
- The gradients will be different as well.
- You need to introduce a tolerance input to `GradientDescent` function that makes the for loop stop when the parameter values converge.

After you complete all your functions, write the main `logistic_regression` function. Remember that it needs to initialize the parameter values, optimize them, (train the ML model) predict the test labels and then, evaluate the test performance. **Keep in mind that the evaluation metric for classification is accuracy.** The `linear_regression` function we wrote in class can be helpful.

Lastly, run the logistic regression function as instructed in the notebook. Tune the stepsize, tolerance and number of iterations. **You need to match the accuracy we have achieved in class to get full credit.**

After you complete and save your code, please submit your .ipynb file via Canvas.